

Q1. I am a regular reader of EFY since 1997. Please briefly explain the following:

1. RoHS technology
2. ESD safe zone

Also, What is the main function of antistatic apron?

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A1. **RoHS.** RoHS stands for restriction of (certain) hazardous substances. It is not a technology but a directive, which aims to restrict the use of certain dangerous substances commonly used in electrical and electronics equipment.



Fig. 1: RoHS compliant logos



Fig. 2: Static safe area notice

It is closely linked with the waste electrical and electronic equipment (WEEE) directive 2002/96/EC, which sets collection, recycling and recovery targets for electrical goods, and is part of a legislative initiative to solve the problem of large amounts of toxic e-waste. The WEEE directive imposes the responsibility for the disposal of waste electrical and electronics equipment on the manufacturers.

The RoHS directive is often referred to as the lead-free directive, but it restricts the use of the following six hazardous materials during the manufacturing of various types of electronics and electrical equipment:

1. Lead (Pb)
2. Mercury (Hg)
3. Cadmium (Cd)
4. Hexavalent chromium (Cr6+)
5. Polybrominated biphenyls (PBB)
6. Polybrominated diphenyl ether (PBDE)

The directive applies to equipment falling in the following categories, as defined by a section of the WEEE

directive:

1. Large and small household appliances
2. IT equipment
3. Telecommunication equipment
4. Consumer equipment
5. Lighting equipment including light bulbs
6. Electronic and electrical tools
7. Toys, leisure and sports equipment
8. Medical devices (currently exempted)
9. Monitoring and control instruments (currently exempted)
10. Automatic dispensers

Equipment complying with RoHS directive carries a logo. Two logos are given in Fig. 1.

ESD safe zone. Electrostatic discharge or ESD is a single-event rapid transfer of electrostatic charge between two objects. It usually results when two objects at different potentials come into direct contact with each other. ESD can also occur when a high electrostatic field develops between two objects in close proximity.

ESD is one of the major causes of device failures in the semiconductor industry. Number of steps are taken to prevent damage from ESD. One of them is the creation of a safe area/zone. Generally, organisations handling and producing static-sensitive parts designate certain areas as 'ESD safe areas' or 'static free zones.'

An ESD safe area requires vigilant compliance with personnel grounding procedures as well as numerous other protocols for preventing the generation and accumulation of static electricity. The processes in these environments are usually governed by international industry standards like ANSI/ESD S20.20 and IEC 61340-5-1. Static safe areas have sign posted as shown in Fig. 2.

A person working in a static-safe work area is supposed to wear an antistatic apron. Other personnel clothing includes gloves, caps, jump suit, wrist bands, shoes, etc. Use of such protective items prevents ESD from persons

to electronic components and systems.

Q2. How can I calculate power factor of a load if I know its inductance? If an inductive load was put to a DC, what is the power factor?

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A2. Power factor is the ratio of true power (watts) to apparent power (volt amps). These are identical only when current and voltage are in phase. If these are in non-reactive load the power factor is 1.0. The power in an AC circuit is very seldom equal to the direct product of volts and amperes.

Average power in an AC circuit expressed in terms of the root mean square voltage (V), current (I) and phase angle (ϕ) between them is:

$$P_{avg} = VI \cos \phi$$

The phasor diagram (Fig. 3) depicts the relationship between these. Here, $R/Z = \cos \phi$, is the power factor and X is the reactive component due to inductive load.

For a DC circuit the power is:

$P = VI = I^2 R$, as V and I are in the same phase and $\cos \phi = 1$.

In terms of power relationship:

Power factor = true power/apparent power

Industrial facilities have large number of induction motors and therefore tend to have a 'lagging power factor,' where the current lags the voltage and power factor is less than one. The electric company has to supply more current to the user for a given amount of power use. Users with inductive loads, therefore have large banks of capacitors installed for the improvement of power factor to avoid penalty from electric company.

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